

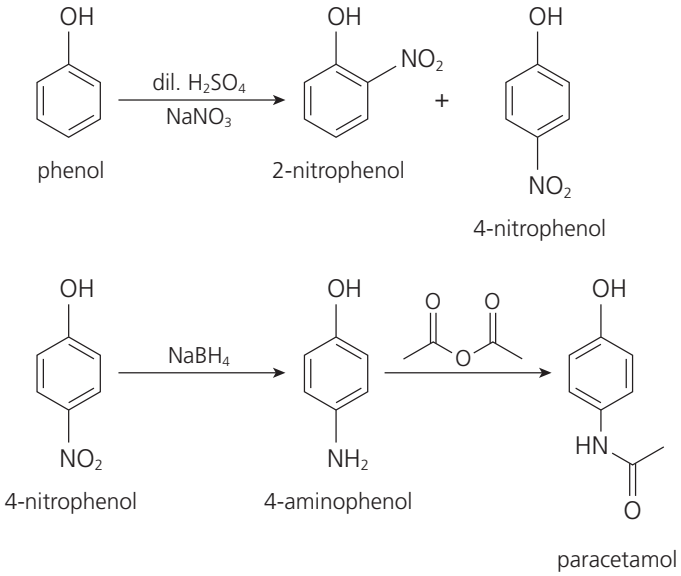
Topic 8

Chemistry of Carbon Compounds

Chemistry in Context

Drug synthesis Synthesis of carbon compounds is a very important branch of chemistry. Scientists, particularly in the pharmaceutical industry, construct carbon compounds via different reactions to make new drugs to cure various diseases. The synthetic route on the right shows the conversion of phenol to paracetamol, a common painkiller.

Researches on syntheses of carbon compounds have led to the development of many useful products such as detergents, nylon and polyesters.



Do you recall?

1 What functional group does phenol contain?
Hydroxyl group

2 What is the molecular formula of paracetamol?
$C_8H_9NO_2$

3 What type of reaction is involved in the second step of the above conversion?
Reduction

In this topic, you are going to study the reactions of carbon compounds of different homologous series, and then plan the synthetic routes for simple carbon compounds.

Unit

29 Naming and physical properties of carbon compounds

Unit Key Concepts

- Introduction to selected homologous series
- Systematic naming of carbon compounds
- Physical properties of carbon compounds

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

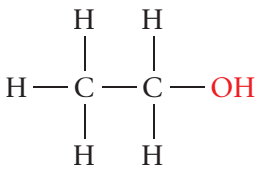
29.1 Functional groups: centre of reactivity

Functional groups provide the key to carbon compounds. A knowledge of the properties and reactions of a limited number of functional groups has opened up our understanding for most carbon compounds.

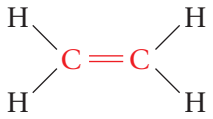
✓ A functional group is an atom, or a group of atoms, which determines most of the properties of a compound.

We have discussed these in Topic 7 Fossil Fuels and Carbon Compounds.

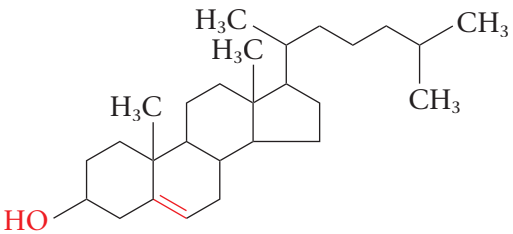
Chemically, a functional group behaves in nearly the same way in every molecule. For example, the hydroxyl group –OH and the carbon-carbon double bond C=C react in characteristic ways, whether they are found in simple molecules such as ethanol and ethene or in more complex molecules such as *cholesterol*. Thus, it is possible to predict the reactions of complex molecules based on the reactions of simple molecules with the same functional groups.



ethanol



ethene



cholesterol

The letter R denotes an alkyl group (C_nH_{2n+1}–) or an aryl group (a group obtained by removing a hydrogen atom attached to a carbon atom in a ring of an aromatic hydrocarbon).

Compounds containing the same functional group form a **homologous series**. Table 29.1 shows some common functional groups and homologous series.

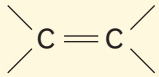
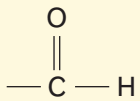
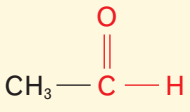
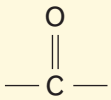
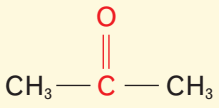
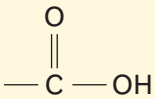
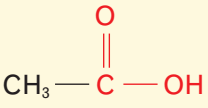
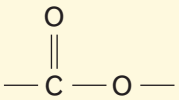
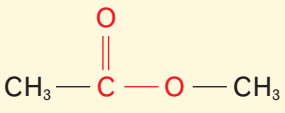
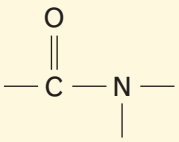
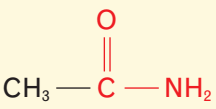
functional group 官能基　　cholesterol 膽固醇　　homologous series 同系列

ASL 2007 Paper 1 Q8(b)
ASL 2009 Paper 1 Q3(c)
AL 2007 Paper 1 Q6(b)
AL 2009 Paper 2 Q5(a)(i)

N1 Refer to page T19.

Table 29.1

Some common functional groups and homologous series

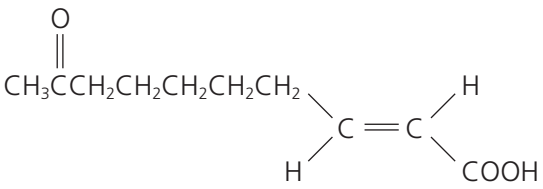
Functional group N1	Homologous series		Example
Formula	General formula	Name	
	C_nH_{2n} (*acyclic*)	alkenes	$H_2C=CH_2$ ethene
–X (X = F, Cl, Br, I)	RX	haloalkanes	CH_3CH_2Cl chloroethane
–OH	ROH	alcohols	CH_3CH_2OH ethanol
	(H or R)CHO	aldehydes	 ethanal
	$RCOR_1$	ketones	 propanone
	(H or R)COOH	carboxylic acids	 ethanoic acid
	$(H \text{ or } R)COOR_1$	esters	 methyl ethanoate
	(H or R)$CONH_2$ (unsubstituted amide)	amides	 ethanamide
	RNH_2 (primary amine)	amines	$CH_3CH_2NH_2$ ethanamine

acyclic 無環的

Practice 29.1

Insects produce pheromones to attract mates or to give an alarm signal. The structure of the pheromone produced by queen bees is shown below. Identify the functional groups in the structure.

Carbonyl group or C=O group
Carbon-carbon double bond
Carboxyl group



29.1

Building molecular models of compounds in different homologous series.

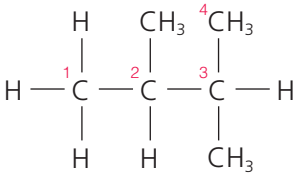
29.2 Naming alkanes and alkenes

In Topic 7 Fossil Fuels and Carbon Compounds, we learnt how to name **alkanes** and **alkenes**. Do the following exercise to remind yourself of how to name alkanes and alkenes using the IUPAC (*International Union of Pure and Applied Chemistry*) system.

Practice 29.2

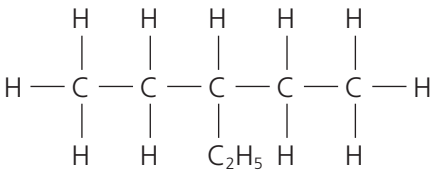
Give the IUPAC names of the following compounds:

a)



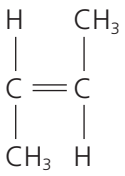
2,3-dimethylbutane

b)



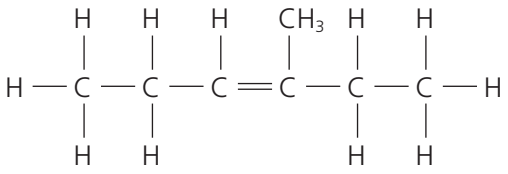
3-ethylpentane

c)



but-2-ene

d)

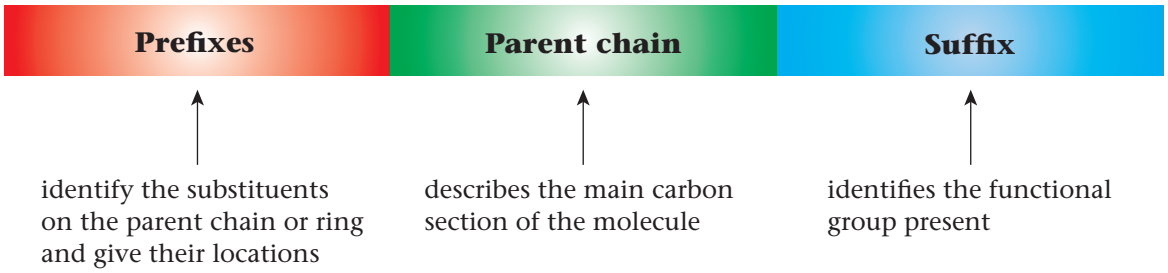


3-methylhex-3-ene

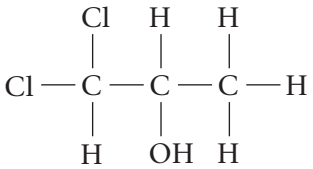
alkane 烷　alkene 烯　International Union of Pure and Applied Chemistry 國際純粹與應用化學聯合會

29.3 The IUPAC rules of naming carbon compounds

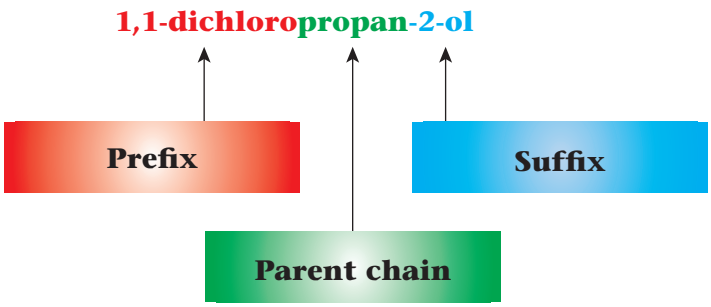
Carbon compounds are divided into several different homologous series. The name of each compound will vary depending on which series it is in. In general, each name has the following components:



For example, consider the following compound:



The IUPAC name of the compound is



The IUPAC rules of naming are as follows:

Table 29.2

The IUPAC rules of naming

Step	Detail
1 Determine the name of the parent chain.	a) Locate the longest continuous chain of carbon atoms containing the carbon bearing the functional group as the parent chain. b) Name the parent chain according to the number of carbon atoms it contains. (Refer to Table 29.3 for the list of names of parent chains.) c) If the parent chain is cyclic, precede the name of the parent chain with the prefix 'cyclo'.
2 Assign a number to each carbon atom in the parent chain.	When naming a molecule other than an alkane, number the parent chain so as to give the lowest number to the carbon atom bearing the functional group.
3 Name the suffix.	a) Add the suffix (showing the functional group) with its position number to the parent chain. b) If two or more identical functional groups are present, give the suffix a numerical prefix according to the number present. (Refer to Table 29.4 for the list of numerical prefixes.)
4 Name the prefixes.	a) Identify each substituent present and assign the name (prefix). (Refer to Table 29.5 for names of substituents.) b) If two or more identical substituents are present, give the substituent name a numerical prefix according to the number present. (Refer to Table 29.4 for the list of numerical prefixes.) c) Assign a number to each substituent, matching the number of the carbon atom where it attaches to the parent chain. d) List the substituent names in alphabetical order◀.

◀ Notice that when deciding the alphabetical order of the prefixes, we ignore the numerical prefixes, such as 'di' and 'tri'.

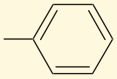
Table 29.3

Names of parent chains			
Number of carbon atom(s)	**Name of parent chain**	**Number of carbon atoms**	**Name of parent chain**
1	meth-	5	pent-
2	eth-	6	hex-
3	prop-	7	hept-
4	but-	8	oct-

Table 29.4

Numerical prefixes	
Number of identical groups	**Numerical prefix**
2	di-
3	tri-
4	tetra-

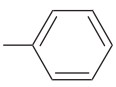
Table 29.5

Names of substituents (always cited as prefixes)	
Substituent	**Name (prefix)**
$-Cl$	chloro-
$-Br$	bromo-
$-I$	iodo-
$-F$	fluoro-
$-CH_3$	methyl
$-CH_2CH_3$	ethyl
$-CH_2CH_2CH_3$	propyl
$-CH_2CH_2CH_2CH_3$	butyl
	phenyl

The structure of benzene is as follows:

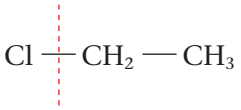


When a benzene ring is attached to other atoms as a substituent, it is called a phenyl group and represented as:

 or $-C_6H_5$

29.4 Naming haloalkanes

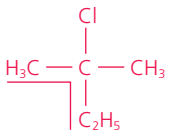
From the study of the alkane series, we know that alkanes can undergo substitution reactions with halogens. If one or more hydrogen atom(s) of an alkane is (are) replaced by the halogen atom(s), the compound is called a **haloalkane**. Chloroethane is a typical haloalkane.



chloroethane

The haloalkanes contain the halogeno functional group: –F, –Cl, –Br or –I. The name of a haloalkane is derived from that of the parent alkane. Every halogen atom is added as a prefix (fluoro-, chloro-, bromo- or iodo-) to the name of the parent alkane. A number is used, if necessary, to indicate the position of every halogen atom in the molecule. Number the parent chain so as to give the lowest number(s) to the carbon atom(s) bearing the substituent(s).

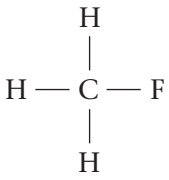
N2 Notice that the alkane parent chain may not appear in a straight line. e.g.



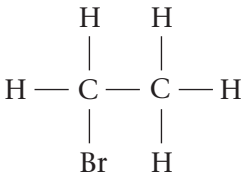
butane parent chain

The IUPAC name of the compound is 2-chloro-2-methylbutane.

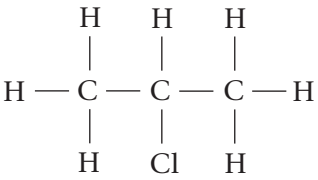
Examples:



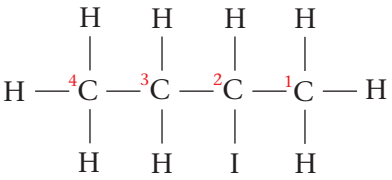
fluoromethane



bromoethane



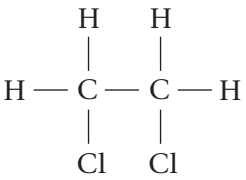
2-chloropropane



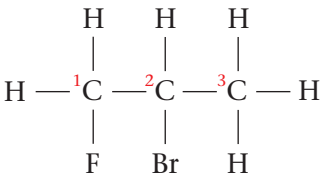
2-iodobutane

We use the prefixes di-, tri-, etc. to show how many atoms of a halogen are present. If there is more than one type of halogen atom, list them in alphabetical order.

Examples:



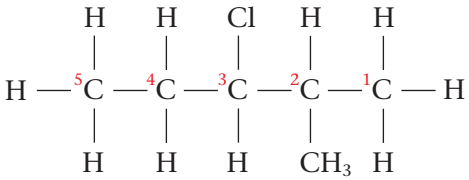
1,2-dichloroethane



2-bromo-1-fluoropropane

haloalkane 卤烷

When the parent chain has both a halogen and an alkyl substituent, number the chain from the end nearest to the first substituent.

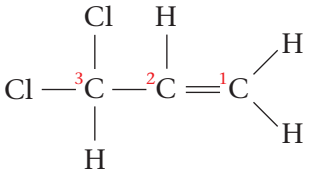


3-chloro-2-methylpentane

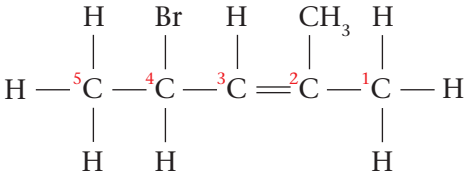
We name an alkene containing halogen atom(s) in a similar way. Remember to number the parent chain starting from the end nearest to the carbon-carbon double bond.

Examples:

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3,3-dichloropropene



4-bromo-2-methylpent-2-ene

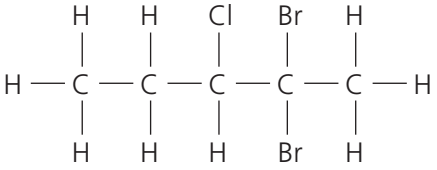
Practice 29.3

1 Give the IUPAC names of the following compounds:

a) $CHCl_3$
trichloromethane

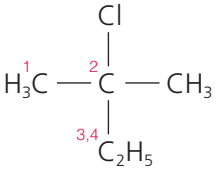
b) $CH_3CHClCH_2CH_2CH_2CH_3$
2-chlorohexane

c)



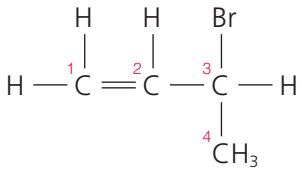
2,2-dibromo-3-chloropentane

d)



2-chloro-2-methylbutane

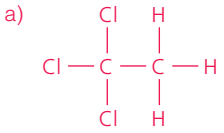
e)



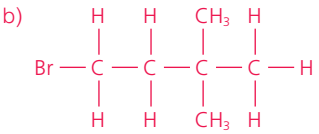
3-bromobut-1-ene

2 Write the structural formulae of the following compounds:

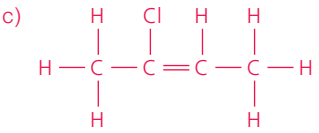
a) 1,1,1-trichloroethane



b) 1-bromo-3,3-dimethylbutane

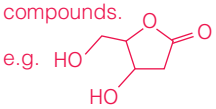


c) 2-chlorobut-2-ene



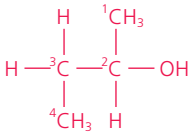
N3 Empirical formula or molecular formula of unfamiliar carbon compounds

Examination questions may ask about the empirical formula or molecular formula of unfamiliar carbon compounds.

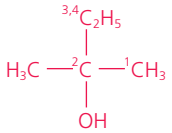
e.g.  molecular formula is $C_5H_8O_4$

N4 Notice that the longest continuous chain containing the carbon bearing the –OH group may not appear in a straight line. Examination questions often ask students to name such compounds.

e.g.



butan-2-ol



2-methylbutan-2-ol

29.5 Naming alcohols

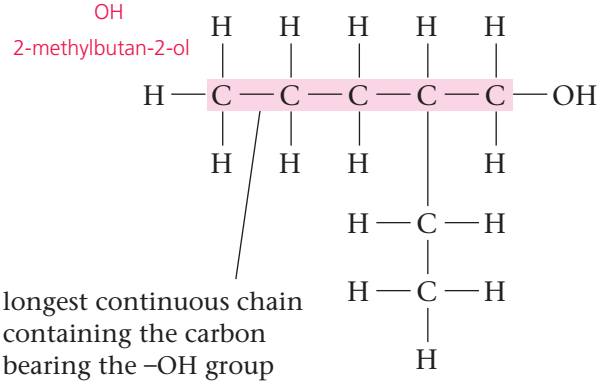
Alcohols are carbon compounds containing one or more **hydroxyl group(s)** (–OH).

In Topic 7 Fossil Fuels and Carbon Compounds, we have learnt how to name alcohols with the general formula $C_nH_{2n+1}OH$ where $n = 1, 2, 3, 4$, etc.. **N3**

A simple alcohol is named as a derivative of the corresponding alkane. The last letter 'e' in the name of the alkane is replaced by the suffix -ol.

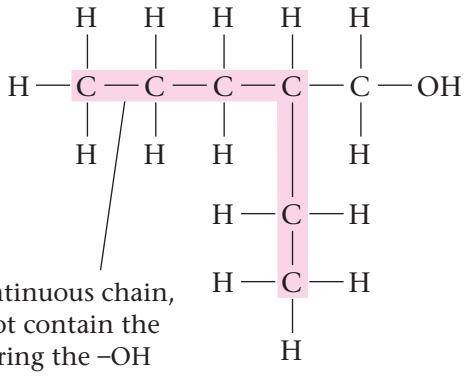
In more complicated, branched systems, the name of the alcohol is based on the longest continuous chain containing the carbon bearing the –OH group. Notice that this chain is not necessarily the longest continuous chain in the molecule. Remember to number the parent chain starting from the end nearest to the –OH group. **N4**

Example:



longest continuous chain containing the carbon bearing the –OH group

✓



longest continuous chain, but does not contain the carbon bearing the –OH group

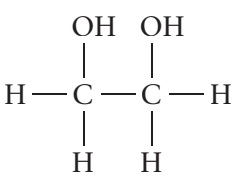
✗

The name of this alcohol is 2-ethylpentan-1-ol.

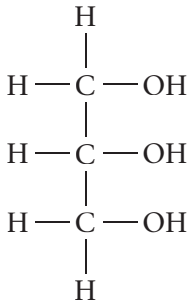
If there is more than one –OH group we use di-, tri-, etc. to say how many –OH groups there are and numbers to say where they are.

alcohol 醇 hydroxyl group 羥基

Examples:



ethane-1,2-diol



propane-1,2,3-triol

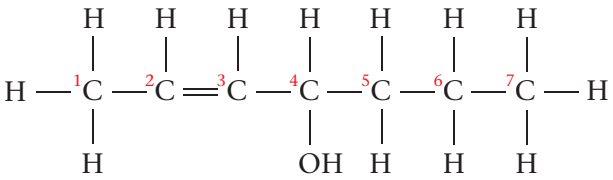
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Notice that the last letter 'e' in the name of the corresponding alkanes are retained in diols and triols.

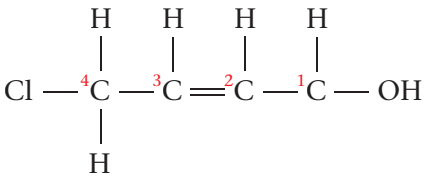
N5 Refer to page T20.

For an alcohol with a C=C bond, add the suffix -en- and a number to indicate the position of the C=C bond. Put these in front of the N5 suffix -ol.

Examples:



hept-2-en-4-ol

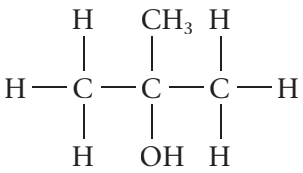


4-chlorobut-2-en-1-ol

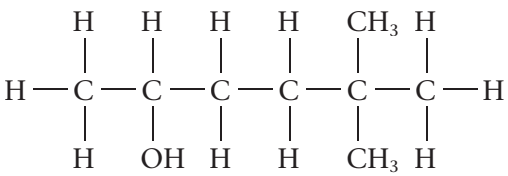
Example 29.1

Q Name the following compounds:

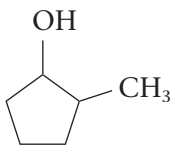
a)



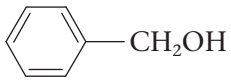
b)



c)

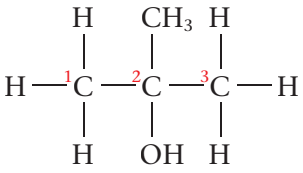


d)



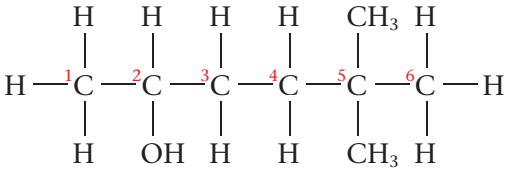
Continued on next page

A1 a)



methylpropan-2-ol

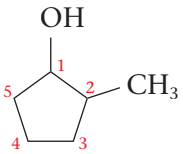
b)



5,5-dimethylhexan-2-ol

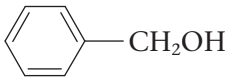
For the compound in part (a), there is only one possible position for the methyl group. So, it is not necessary to assign a number to indicate its position.

c)



2-methylcyclopentanol

d)

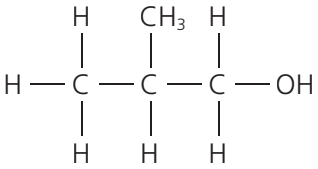


phenylmethanol

Practice 29.4

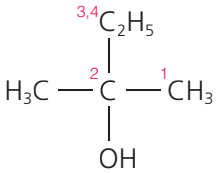
1 Give the IUPAC names of the following compounds:

a)



methylpropan-1-ol

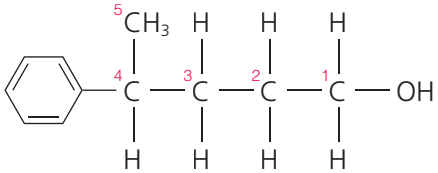
b)



2-methylbutan-2-ol

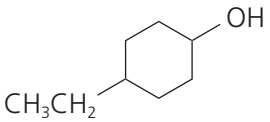
CE 2007 Paper 2 Q8

c)



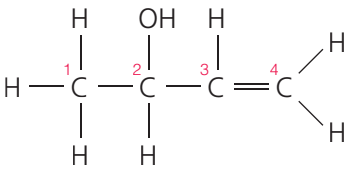
4-phenylpentan-1-ol

d)



4-ethylcyclohexanol

e)



but-3-en-2-ol

2 Write the structural formulae of the following compounds: Refer to the next page.

a) hexan-3-ol

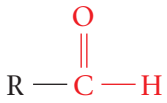
b) 2-phenylethanol

c) butane-2,3-diol

d) pent-1-en-3-ol

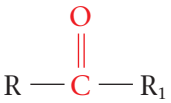
29.6 Naming aldehydes and ketones

Aldehydes and **ketones** both contain the **carbonyl group**,  , but they differ in the position of the carbonyl group in the hydrocarbon skeleton. Aldehydes have the carbonyl group in the terminal position of the chain. Their general formula is therefore RCHO, where R is an alkyl or aryl group or hydrogen. Ketones have the carbonyl group in a non-terminal position of the chain. The general formula is $RCOR_1$ where R and R_1 are alkyl or aryl groups.



aldehyde

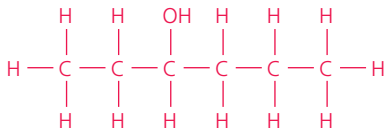
where R = an alkyl or aryl group or H

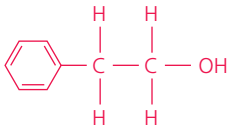


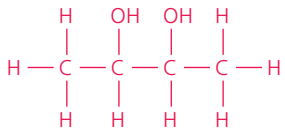
ketone

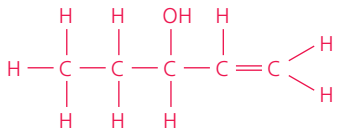
where R and R_1 = alkyl or aryl groups

Practice 29.4

2 a) 

b) 

c) 

d) 

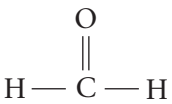
Thus, aldehydes and ketones are structurally quite similar. However, their properties differ considerably and they are considered as different homologous series.

Naming aldehydes

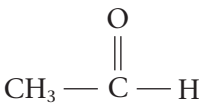
An aldehyde is named by replacing the last letter 'e' in the name of the corresponding alkane with the suffix -al.

Since the  group is always in the terminal position of the carbon chain, there is no need to indicate its position. Remember that the carbon atom of the carbonyl group is assigned number 1.

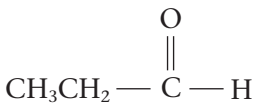
Examples:



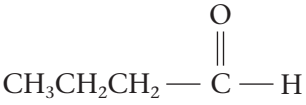
methanal



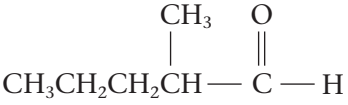
ethanal



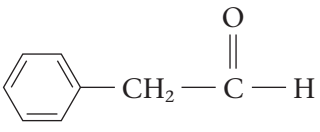
propanal



butanal



2-methylpentanal

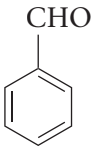


phenylethanal

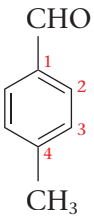
aldehyde 醛　ketone 酮　carbonyl group 羰基

Aldehydes in which the  group is directly attached to a benzene ring are benzaldehydes.

Examples:



benzaldehyde



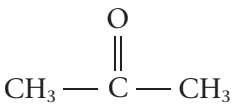
4-methylbenzaldehyde

Naming ketones

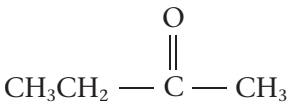
A ketone is named by replacing the last letter 'e' in the name of the corresponding alkane with the suffix -one.

The common name of propanone is acetone.

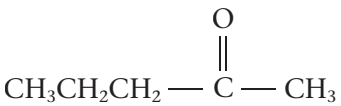
Examples:



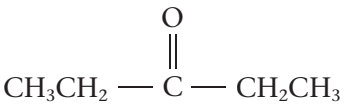
propanone



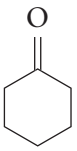
butanone



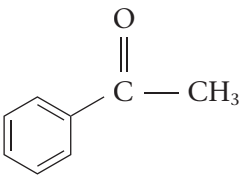
pentan-2-one



pentan-3-one



cyclohexanone

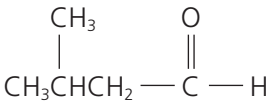


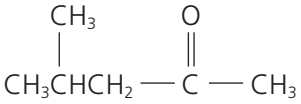
phenylethanone

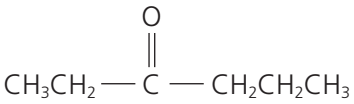
We do not need to specify the position of the carbonyl group in propanone or butanone because the carbonyl group can only be in one position. For ketones with more carbon atoms, we specify the position of the carbonyl group by inserting the number of the carbon atom bearing the group. Thus, $CH_3CH_2CH_2COCH_3$ is called pentan-2-one while $CH_3CH_2COCH_2CH_3$ is called pentan-3-one.

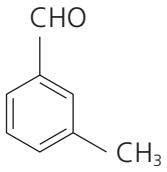
Practice 29.5

1 Give the IUPAC names of the following compounds:

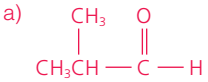
a) 
3-methylbutanal

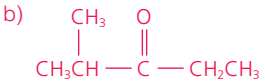
b) 
4-methylpentan-2-one

c) 
hexan-3-one

d) 
3-methylbenzaldehyde

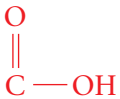
2 Write the structural formulae of the following compounds:

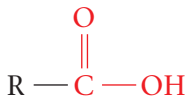
a) methylpropanal 

b) 2-methylpentan-3-one 

c) cyclohexanone 

29.7 Naming carboxylic acids

Carboxylic acids contain the **carboxyl group**,  . Their general formula is usually written as RCOOH, where R is an alkyl or aryl group or hydrogen.



where R = an alkyl or aryl group or H

A carboxylic acid is named by replacing the last letter 'e' in the name of the corresponding alkane with the suffix -oic acid. Remember that the carbon atom of the carboxyl group is assigned number 1.

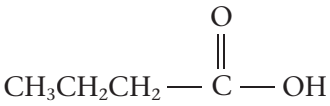
carboxylic acid 羧酸　　carboxyl group 羧基

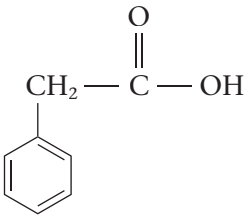
Examples:

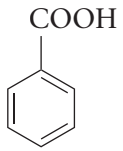

methanoic acid


ethanoic acid


propanoic acid


butanoic acid


phenylethanoic acid

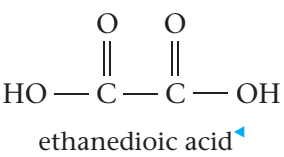

benzoic acid

Notice that the last letter 'e' in the name of the corresponding alkanes are retained in dioic acids.

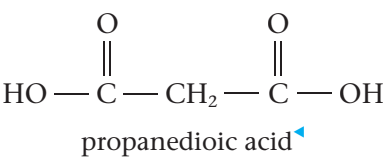
There is no need to assign numbers to indicate the positions of the carboxyl groups as they must be at terminal positions.

Where two carboxyl groups are present, the acid is dibasic and the suffix -dioic acid is used.

Examples:

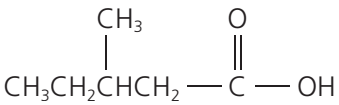

ethanedioic acid

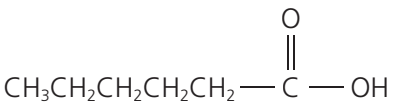
DSE 2013 Paper 1B Q4(a)

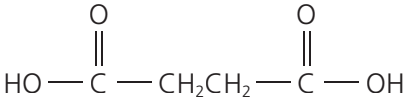

propanedioic acid

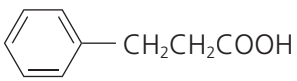
Practice 29.6

1 Give the IUPAC names of the following compounds:

a) 
3-methylpentanoic acid

b) 
hexanoic acid

c) 
butanedioic acid

d) 
3-phenylpropanoic acid

2 Write the structural formulae of the following compounds: Refer to the next page.

a) 3,3-dimethylbutanoic acid

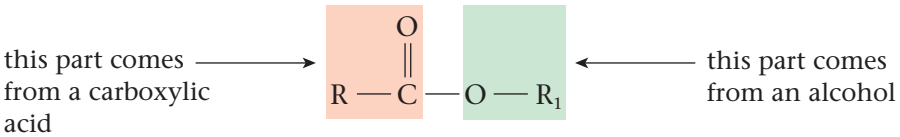
b) hexanedioic acid

c) 4-methylbenzoic acid

29.8 Naming esters

Esters are formed when an alcohol reacts with a carboxylic acid. They contain the ester functional group .

Esters have a general formula $RCOOR_1$, where R is an alkyl or aryl group or hydrogen, and R_1 is an alkyl or aryl group.

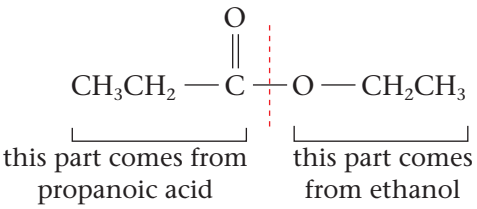


where R = an alkyl or aryl group or H
R_1 = an alkyl or aryl group

The name of an ester consists of two separate words. The first word of the name indicates the alcohol from which the ester is derived, with the suffix -yl. The second word indicates the acid from which the ester is derived, with the '-oic acid' ending replaced by -oate.

We can work out the name of an ester easily from its structural formula as shown below:

1 Divide the formula into two portions by mentally drawing a line before the bridging O of the –COO group, e.g.

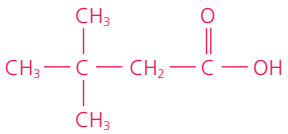


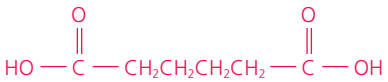
2 Name the part that comes from ethanol, i.e. *ethyl*.

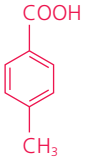
3 Name the part that comes from propanoic acid, i.e. *propanoate*.

4 Combine the two to give the name of the ester — *ethyl propanoate*.

Practice 29.6

2 a) 

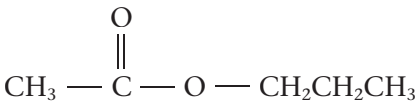
b) 

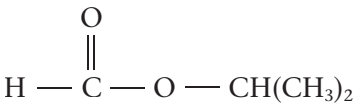
c) 

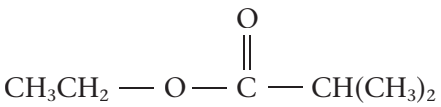
ester 酯

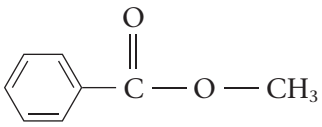
Example 29.2

Q Give the IUPAC names of the following esters:

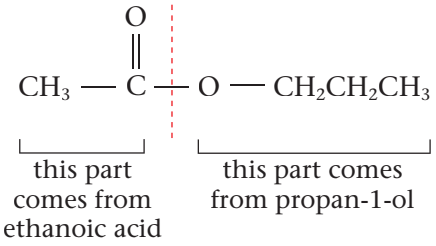
a) 

b) 

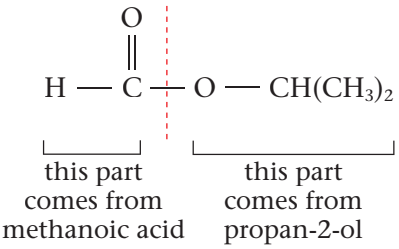
c) 

d) 

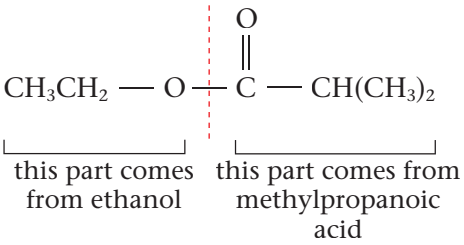
Refer to part (c). We may write the alcohol part first in the structural formula of an ester.

A a) 

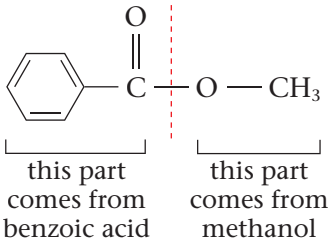
1-propyl ethanoate

b) 

2-propyl methanoate

c) 

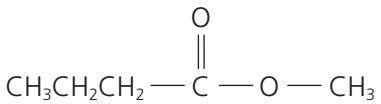
ethyl methylpropanoate

d) 

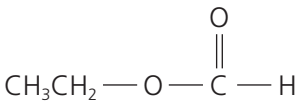
methyl benzoate

Practice 29.7

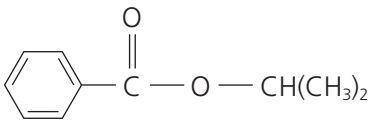
1 Give the IUPAC names of the following compounds:

a) 

methyl butanoate

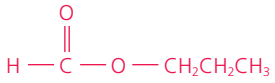
b) 

ethyl methanoate

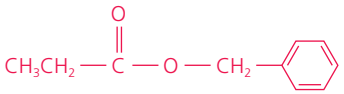
c) 

2-propyl benzoate

2 Write the structural formulae of the following compounds:

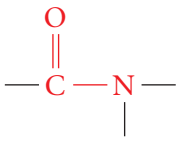
a) 1-propyl methanoate 

b) phenylmethyl propanoate

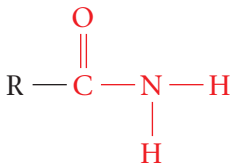
b) 

29.9 Naming amides

Amides are nitrogen compounds derived from carboxylic acids. They contain the amide functional group:



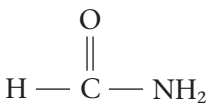
Amides with an unsubstituted $-NH_2$ group have the general formula:



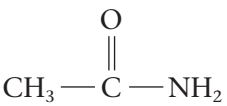
where R = an alkyl or aryl group or H

An amide is named by replacing the 'oic acid' ending in the name of the corresponding carboxylic acid with -amide.

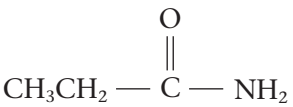
Examples:



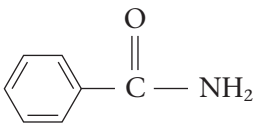
methanamide



ethanamide



propanamide



benzamide

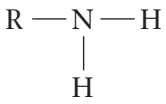
amide 酰胺

29.10 Naming amines

We can consider **amines** as derivatives of ammonia in which one or more of the hydrogen atoms in the ammonia molecule have been replaced by alkyl or aryl groups.

Amines are classified as primary (1°), secondary (2°) or tertiary (3°), depending on how many substituents are connected to the nitrogen atom. We will focus on primary amines at this level.

Primary amines has the general formula:



where R = an alkyl or aryl group

A primary amine is named by replacing the last letter 'e' in the name of the corresponding alkane with the suffix -amine.

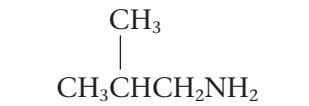
Examples:



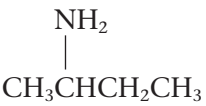
methanamine



ethanamine



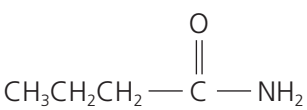
2-methylpropan-1-amine



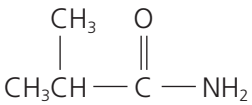
butan-2-amine

Practice 29.8

1 Give the IUPAC names of the following compounds:

a) 

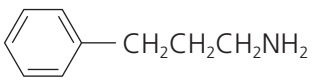
butanamide

b) 

methylpropanamide

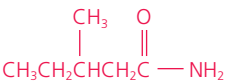
c) $CH_3CH_2CH_2CH_2NH_2$

butan-1-amine

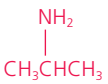
d) 

3-phenylpropan-1-amine

2 Write the structural formulae of the following compounds:

a) 3-methylpentanamide 

b) propan-2-amine



amine 胺

29.11 Intermolecular forces and physical properties of carbon compounds

The physical properties of a carbon compound are related to the type of forces between its molecules, such as van der Waals' forces or hydrogen bonds.

The physical properties of a carbon compound therefore depend to a certain extent on the functional group it contains, as the group determines the type of intermolecular forces that is present. Moreover, the strength of intermolecular forces is affected by the length of the carbon chain in molecules. In the following sections, we shall examine how these two factors affect the physical properties of carbon compounds.

Do you know?

Chlorofluorocarbons (CFCs)

Chlorofluorocarbons (CFCs) are haloalkanes containing chlorine and fluorine but no hydrogen. At room temperature, they are usually colourless gases or liquids (b.p. ~30 °C) which evaporate easily. They are examples of volatile organic compounds (VOCs). They are stable, non-toxic and non-flammable.

The properties of CFCs make them useful for a variety of commercial and industrial purpose: as a propellant in aerosol sprays (now banned in the US and EU), in refrigeration and air-conditioning systems, in foams, in cleansing solvents and electrical components.

Most CFCs have been released to the atmosphere from aerosols and leakages of refrigeration equipment. Their long lifetimes in the atmosphere mean that some end up in the higher atmosphere. They can then destroy the ozone layer, thus reducing the protection it offers the earth from the sun's harmful UV rays.

29.12 Physical properties of haloalkanes

Haloalkanes have a C–X bond. This bond is polar, $\overset{\delta+}{C}-\overset{\delta-}{X}$, because halogens are more electronegative than carbon.

Boiling point

The intermolecular forces in haloalkanes are van der Waals' forces.

The boiling point of a haloalkane depends on the length of the carbon chain in its molecules and what the halogen is.

- Boiling point increases with increased length of the carbon chain.
- Boiling point increases as we go down the halogen group.

Both these effects are caused by increased van der Waals' forces because the larger a molecule is, the greater is the number of electrons and hence the stronger the van der Waals' forces. More heat is needed to separate the molecules during boiling and thus the boiling point increases.

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Most haloalkanes, except some simple ones which are gases, are liquids at room temperature. Table 29.6 lists the boiling points of some haloalkanes.

Table 29.6

Boiling points of some haloalkanes

Alkyl group / IUPAC name \ Halogen	Boiling point (°C)					
	Cl		Br		I	
CH_3–	chloromethane	–24	bromomethane	4	iodomethane	43
C_2H_5–	chloroethane	13	bromoethane	38	iodoethane	72
C_3H_7–	1-chloropropane	47	1-bromopropane	71	1-iodopropane	102
C_4H_9–	1-chlorobutane	79	1-bromobutane	101	1-iodobutane	130

Haloalkanes have higher boiling points than alkanes of similar chain lengths because they are heavier and polar.

Solubility in water

The polar haloalkane molecules can interact with water molecules, but the attractive forces set up are not as strong as the hydrogen bonds between water molecules. Hence haloalkanes are only slightly soluble in water.

29.13 Physical properties of alcohols

Table 29.7 lists the boiling points and water solubilities of six alcohols.

'Infinite' means miscible in all proportions.

Table 29.7

Boiling points and water solubilities of six alcohols

Alcohol	Structural formula	Boiling point (°C)	Solubility (g per 100 g of water)
Methanol	CH_3OH	65	infinite
Ethanol	CH_3CH_2OH	78	infinite
Propan-1-ol	$CH_3CH_2CH_2OH$	97	infinite
Butan-1-ol	$CH_3CH_2CH_2CH_2OH$	118	7.9
Pentan-1-ol	$CH_3CH_2CH_2CH_2CH_2OH$	138	2.3
Hexan-1-ol	$CH_3CH_2CH_2CH_2CH_2CH_2OH$	157	0.6

Boiling point

We have discussed hydrogen bonds between molecules in Topic 6 Microscopic World II.

Two types of intermolecular forces — hydrogen bonds and van der Waals' forces — operate between alcohol molecules.

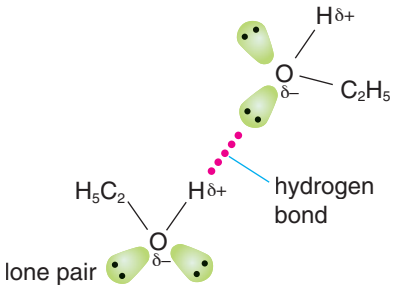


Fig. 29.1 Hydrogen bonding between ethanol molecules

All the alcohols in Table 29.7 have one –OH group for hydrogen bond formation (Fig. 29.1). Their boiling point difference is due to the difference in the strength of van der Waals' forces between the molecules.

The strength of van der Waals' forces increases with the length of the carbon chain. More heat is needed to separate the larger molecules during boiling. Thus, the boiling points of the alcohols increase with the length of the carbon chain. The boiling points of alcohols are much higher than those of alkanes of similar relative molecular masses (Table 29.8).

ASL 2005 Q9(a)(i)

Table 29.8

Boiling points of alcohols and alkanes of similar relative molecular masses

Alcohol			Alkane		
Structural formula	**Relative molecular mass**	**Boiling point (°C)**	**Structural formula**	**Relative molecular mass**	**Boiling point (°C)**
CH_3OH	32	65	CH_3CH_3	30	–89
CH_3CH_2OH	46	78	$CH_3CH_2CH_3$	44	–42
$CH_3CH_2CH_2OH$	60	97	$CH_3CH_2CH_2CH_3$	58	–0.5
$CH_3CH_2CH_2CH_2OH$	74	118	$CH_3CH_2CH_2CH_2CH_3$	72	36
$CH_3CH_2CH_2CH_2CH_2OH$	88	138	$CH_3CH_2CH_2CH_2CH_2CH_3$	86	69
$CH_3CH_2CH_2CH_2CH_2CH_2OH$	102	157	$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$	100	98

Do you know

Methylated spirits

Methylated spirits (Fig. 29.3) is ethanol mixed with additives to make it poisonous and bitter. The main additive is about 10% of methanol, giving rise to the term 'methylated spirits'. Hence methylated spirits cannot be served as an alcoholic beverage. Drinking methylated spirits lead to blindness or even death.

Methylated spirits can be used as a solvent to clean oil-based paints. It can also be used as glass cleanser and a fuel for camp stoves.



Fig. 29.3 Some methylated spirits is dyed to discourage ingestion

Reference Website

Refer to the following animation for the hydrogen bonding between molecules of ethanol and water:
http://chemmovies.unl.edu/ChemAnime/ETOHWD/ETOHWD.html
(accessed July 2014)

Solubility in water

Due to the hydrogen bonding between alcohol molecules and water molecules (Fig. 29.2), alcohols with less carbon atoms (C_1, C_2, C_3 and some isomers of C_4) are miscible with water in all proportions.

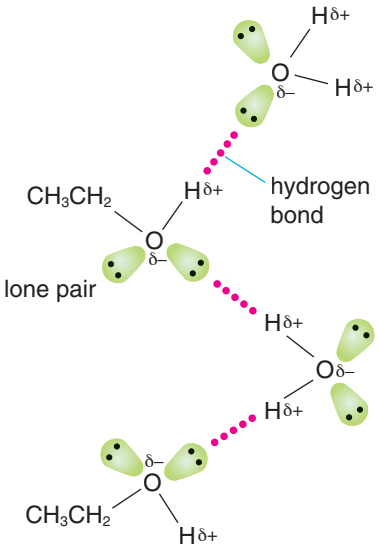


Fig. 29.2 Hydrogen bonding between ethanol molecules and water molecules

A rule of thumb for predicting solubility is that 'like dissolves like'; polar compounds tend to dissolve in polar solvents.

Both methanol and ethanol are miscible with water in all proportions. The alkyl groups of these alcohols are relatively small, therefore these alcohols resemble water more than they resemble alkanes.

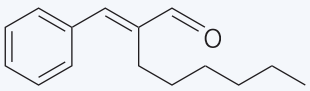
Alcohols with long carbon chains are much less soluble in water. N6 This is because these alcohols resemble alkanes more than they resemble water. Van der Waals' forces predominate between these alcohol molecules.

N6 Ethane-1,2-diol is soluble in water because
- it has a short carbon chain;
- the –OH groups in it can form hydrogen bonds with water.

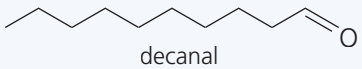
Do you know ?

Chanel No 5

Chanel No 5 is one of the best selling perfumes in the world. It was launched by the French fashion designer Coco Chanel in 1921.The chemical formula for the fragrance was developed by the Russian-French chemist and perfume designer Ernest Beaux. Working on creation of this perfume, Ernest beaux mixed aldehydes with fragrance of rose and jasmine. Many synthetically produced aldehydes are now used in making perfumes. For example, hexyl cinnamaldehyde has a jasmine fragrance and decanal provides a fragrance of violets, roses and oranges. The structures of these aldehydes are shown below.



hexyl cinnamaldehyde



decanal



Fig. 29.4 A bottle of perfume

29.14 Physical properties of aldehydes and ketones

Table 29.9 lists the boiling points and water solubilities of six aldehydes and ketones.

Table 29.9

Boiling points and water solubilities of six aldehydes and ketones

Name	Structural formula	Boiling point (°C)	Solubility (g per 100 g of water)
Aldehyde			
Methanal	$HCHO$	–21	infinite
Ethanal	CH_3CHO	21	infinite
Propanal	CH_3CH_2CHO	49	31
Butanal	$CH_3CH_2CH_2CHO$	76	7.1
Pentanal	$CH_3CH_2CH_2CH_2CHO$	102	1.2
Hexanal	$CH_3CH_2CH_2CH_2CH_2CHO$	131	0.6
Ketones			
Propanone	CH_3COCH_3	56	infinite
Butanone	$CH_3COCH_2CH_3$	80	22
Pentan-2-one	$CH_3COCH_2CH_2CH_3$	102	4.3
Pentan-3-one	$CH_3CH_2COCH_2CH_3$	102	5.0
Hexan-2-one	$CH_3COCH_2CH_2CH_2CH_3$	127	1.8
Hexan-3-one	$CH_3CH_2COCH_2CH_2CH_3$	123	1.5

Boiling point

Methanal is a gas at room temperature. Ethanal boils at 21 °C, so it may be a liquid or gas at room temperature depending on the conditions. Other short-chain aldehydes and ketones are liquids. The boiling points of members in both series increase with the length of the carbon chain (Table 29.9).

Since aldehydes and ketones cannot form hydrogen bonds between their own molecules, they have lower boiling points than the corresponding alcohols.

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Solubility in water

Aldehydes and ketones composed of molecules with small number of carbon atoms — such as methanal, ethanal and propanone — are miscible with water in all proportions, but this miscibility decreases with increasing length of the carbon chain (Table 29.9).

The reason for the solubility is that they can form hydrogen bonds with water molecules (Fig. 29.5).

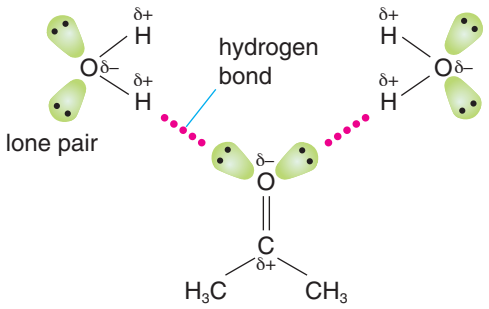


Fig. 29.5 Hydrogen bonding between a propanone molecule and water molecules

29.15 Physical properties of carboxylic acids

Table 29.10 lists the boiling points and water solubilities of six carboxylic acids.

Table 29.10

Boiling points and water solubilities of six carboxylic acids

Carboxylic acid	Structural formula	Boiling point (°C)	Solubility (g per 100 g of water)
Methanoic acid	HCOOH	101	infinite
Ethanoic acid	CH_3COOH	118	infinite
Propanoic acid	CH_3CH_2COOH	141	infinite
Butanoic acid	$CH_3CH_2CH_2COOH$	164	infinite
Pentanoic acid	$CH_3CH_2CH_2CH_2COOH$	187	4.97
Hexanoic acid	$CH_3CH_2CH_2CH_2CH_2COOH$	205	1.08

Boiling point

Most carboxylic acids are liquids at room temperature. In a pure carboxylic acid, hydrogen bonding can form between two acid molecules to produce a **dimer** (Fig. 29.6). This immediately doubles the size of the molecule and so increases the van der Waals' forces between the molecule and its neighbours — resulting in a high boiling point.

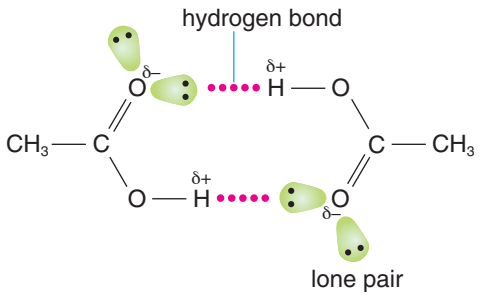


Fig. 29.6 Hydrogen bonding can form between two ethanoic acid molecules to produce a dimer

As the molecules of carboxylic acids can form more extensive hydrogen bonds than those of alcohols do, the boiling points of carboxylic acids are higher than those of alcohols of similar relative molecular masses (Table 29.11).

Table 29.11

Comparing the boiling points of carboxylic acids with alcohols of similar relative molecular masses

Relative molecular masses	Carboxylic acid	Boiling point (°C)	Alcohol	Boiling point (°C)
46	HCOOH	101	CH_3CH_2OH	78
60	CH_3COOH	118	$CH_3CH_2CH_2OH$	97
74	CH_3CH_2COOH	141	$CH_3CH_2CH_2CH_2OH$	118

dimer 二聚物

Fig. 29.7 shows the variation of boiling points of straight-chain carboxylic acids, alcohols, aldehydes, ketones and alkanes with the number of carbon atoms in their molecules.

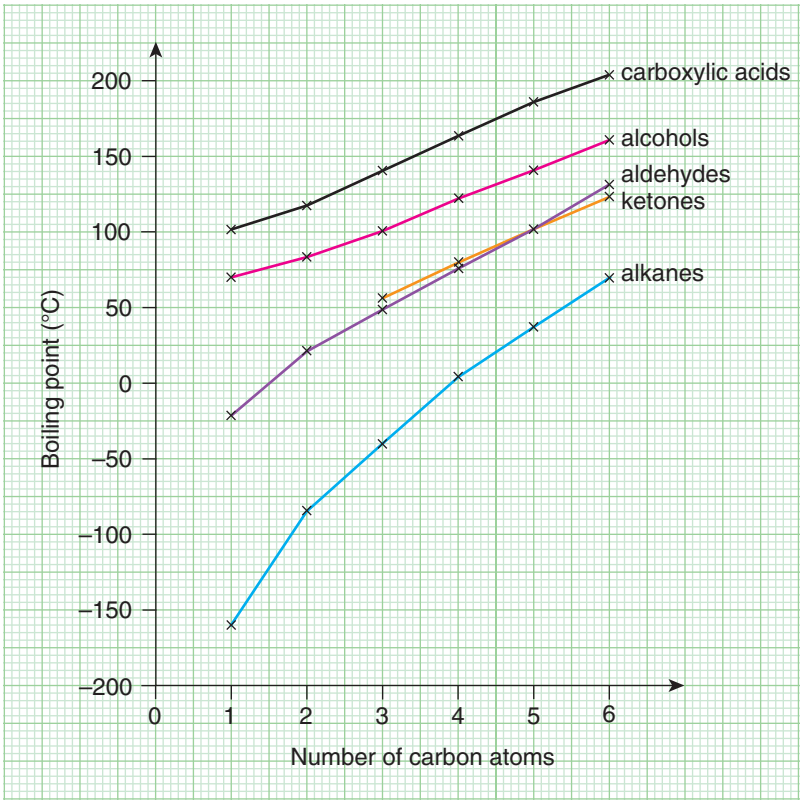


Fig. 29.7 Variation of boiling points of straight-chain carboxylic acids, alcohols, aldehydes, ketones and alkanes with the number of carbon atoms in their molecules

The boiling points of members in the same homologous series increase with the number of carbon atoms in molecules.

Molecules of alcohols can form hydrogen bonds with each other while those of alkanes, aldehydes and ketones cannot. Thus, for the same number of carbon atoms in molecules, the boiling point of alcohol is higher than that of alkane, aldehyde and ketone.

For the same number of carbon atoms in molecules, the boiling point of carboxylic acid is higher than that of alcohol because the acid can form more extensive hydrogen bonds.

Solubility in water

CE 2010 Paper 2 Q18 – The first four carboxylic acids are miscible with water in all proportions because their molecules can form hydrogen bonds with water molecules.

Fig. 29.8 shows that both oxygen atoms and the hydrogen atom of the –OH group of a carboxylic acid molecule can form hydrogen bonds with water molecules.

The water solubility decreases with increasing length of the carbon chain.

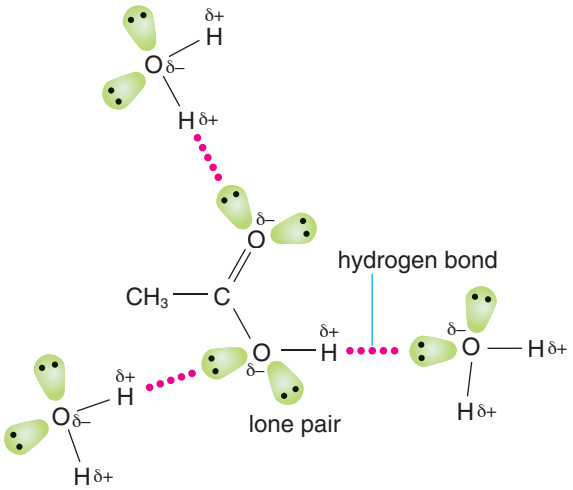


Fig. 29.8 Hydrogen bonding between an ethanoic acid molecule and water molecules

The water solubility of carboxylic acid is higher than that of alcohol with the same number of carbon atoms in the molecules. This is because the molecules of carboxylic acid can form more extensive hydrogen bonds with water molecules.

CE 2009 Paper 2 Q27



Fig. 29.9 The fragrances of fruits are due to a mixture of natural esters

29.16 Physical properties of esters

Esters have pleasant, sweet and fruity smells. The flavours and fragrances of many fruits and flowers are due to a mixture of natural esters (Fig. 29.9).

Perfumes, cosmetics and the *artificial flavourings* present in foods and drinks contain synthetic esters (Table 29.12).

Table 29.12

The fragrances of some synthetic esters	
Ester	**Fragrance**
Ethyl methanoate	raspberry
Ethyl butanoate	pineapple
2-pentyl ethanoate	pear

artificial flavouring 人造香料

Table 29.13 lists the boiling points and water solubilities of eight esters.

Table 29.13

The boiling points and water solubilities of eight esters

Ester	Structural formula	Boiling point (°C)	Solubility (g per 100 g of water)
Methyl methanoate	$HCOOCH_3$	32	30
Ethyl methanoate	$HCOOCH_2CH_3$	54	11
Methyl ethanoate	CH_3COOCH_3	57	24
Ethyl ethanoate	$CH_3COOCH_2CH_3$	77	7.4
Propyl ethanoate	$CH_3COOCH_2CH_2CH_3$	102	1.9
Butyl ethanoate	$CH_3COOCH_2CH_2CH_2CH_3$	125	1.0
Ethyl propanoate	$CH_3CH_2COOCH_2CH_3$	99	1.8
Ethyl butanoate	$CH_3CH_2CH_2COOCH_2CH_3$	120	0.5

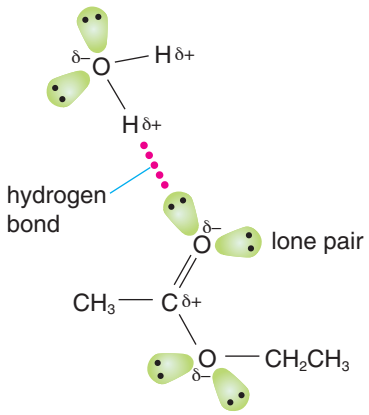


Fig. 29.10 Hydrogen bonding between an ethyl ethanoate molecule and a water molecule

Boiling point

Simple esters are liquids at room temperature. Esters are polar compounds, but, lacking a hydrogen atom attached to any of the two oxygen atoms, their molecules cannot form hydrogen bonds with each other. As a result, the boiling points of esters are lower than those of alcohols and acids of similar relative molecular masses. The boiling points of esters are about the same as those of aldehydes and ketones of similar relative molecular masses.

Solubility in water

CE 2010 Paper 2 Q18

Molecules of esters can form hydrogen bonds with water molecules (Fig. 29.10). As a result, simple esters such as methyl methanoate and methyl ethanoate are very soluble in water. N7

N7 Water solubility data of some simple esters

Ester	Solubility (g per 100 g of water)	Ester	Solubility (g per 100 g of water)
Methyl methanoate	very soluble	Propyl ethanoate	1.9
Ethyl methanoate	soluble	Butyl ethanoate	1.0
Methyl ethanoate	24.4	Ethyl propanoate	1.8
Ethyl ethanoate	7.4	Ethyl butanoate	0.5

29.17 Physical properties of amides

Boiling point

Methanamide is a liquid at room temperature but the other amides are solids. Table 29.14 lists the boiling points of six simple amides.

Table 29.14

The boiling points of six simple amides

Amide	Structural formula	Boiling point (°C)
Methanamide	$HCONH_2$	210
Ethanamide	CH_3CONH_2	221
Propanamide	$CH_3CH_2CONH_2$	213
Butanamide	$CH_3CH_2CH_2CONH_2$	216
Pentanamide	$CH_3CH_2CH_2CH_2CONH_2$	232
Hexanamide	$CH_3CH_2CH_2CH_2CH_2CONH_2$	255

The boiling points of amides are high because they can form extensive hydrogen bonds (Fig. 29.11). Each hydrogen atom in the $-NH_2$ group in a molecule can form a hydrogen bond with a lone pair on the oxygen atom of another molecule.

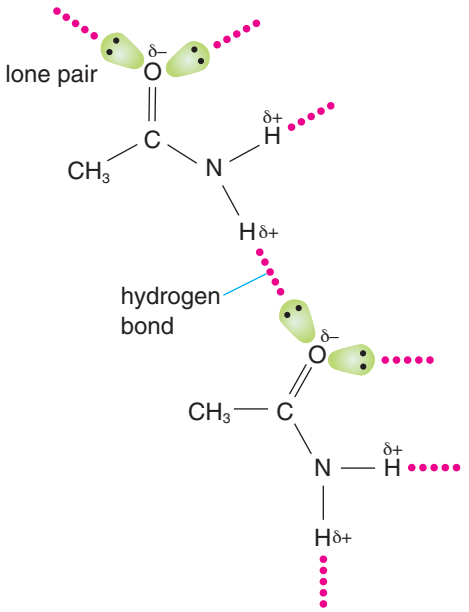


Fig. 29.11 Hydrogen bonding between ethanamide molecules

Solubility in water

Molecules of amides can form hydrogen bonds with water molecules (Fig. 29.12). As a result, simple amides such as methanamide and ethanamide are very soluble in water.

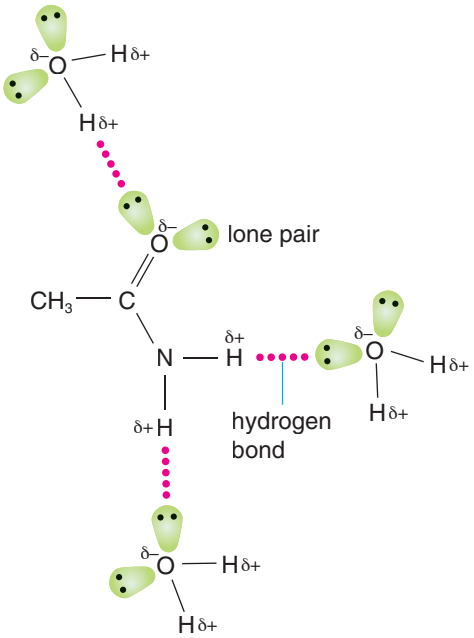


Fig. 29.12 Hydrogen bonding between an ethanamide molecule and water molecules

29.18 Physical properties of amines

The physical properties of the simplest amines are similar to those of ammonia. So, methanamine and ethanamine are gases at room temperature and they smell like ammonia, though with a fishy character (Fig. 29.13).



Fig. 29.13 The fishy smell is partly due to ethanamine

Boiling point

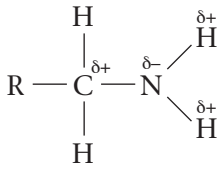
Table 29.15 lists the boiling points of six primary amines.

Table 29.15

The boiling points of six primary amines

Amine	Structural formula	Boiling point (°C)
Methanamine	CH_3NH_2	–6
Ethanamine	$CH_3CH_2NH_2$	17
Propan-1-amine	$CH_3CH_2CH_2NH_2$	49
Butan-1-amine	$CH_3CH_2CH_2CH_2NH_2$	78
Pentan-1-amine	$CH_3CH_2CH_2CH_2CH_2NH_2$	103
Hexan-1-amine	$CH_3CH_2CH_2CH_2CH_2CH_2NH_2$	130

Amines are polar:



Molecules of primary amines can form hydrogen bonds with each other (Fig. 29.14). However, as nitrogen is less electronegative than oxygen, the N–H bond in primary amines is less polar than the O–H bond in alcohols. As a result, the hydrogen bonds in primary amines are not as strong as those in alcohols. This explains why primary amines have lower boiling points than alcohols of similar relative molecular masses.

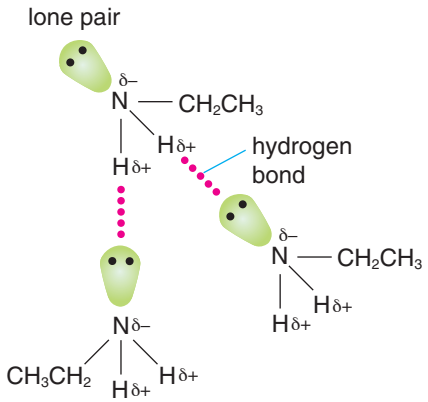


Fig. 29.14 Hydrogen bonding between ethanamine molecules

Table 29.16 shows the comparison of the boiling points of primary amines with alcohols of similar relative molecular masses.

Table 29.16

Comparing the boiling points of primary amines with alcohols of similar relative molecular masses

Structural formula	Relative molecular mass	Boiling point (°C)	Structural formula	Relative molecular mass	Boiling point (°C)
CH_3NH_2	31	–6	CH_3OH	32	65
$CH_3CH_2NH_2$	45	17	CH_3CH_2OH	46	78
$CH_3CH_2CH_2NH_2$	59	49	$CH_3CH_2CH_2OH$	60	97

Solubility in water

Like the ammonia molecule, molecules of primary amines can form hydrogen bonds with water molecules (Fig. 29.15). As a result, simple primary amines such as methanamine, ethanamine, propan-1-amine and butan-1-amine are very soluble in water.

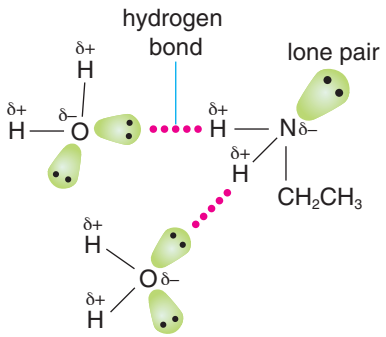


Fig. 29.15 Hydrogen bonding between an ethanamine molecule and water molecules

A solution of an amine in water is alkaline.

For example, when propan-1-amine dissolves in water, hydroxide ions are formed.

$$CH_3CH_2CH_2NH_2(l) + H_2O(l) \rightleftharpoons [CH_3CH_2CH_2NH_3]^+(aq) + OH^-(aq)$$

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Arrange the following compounds in order of increasing boiling point. Explain your answer. N8

CH_3CH_2CHO	$CH_3CH_2CH_2COOH$	$CH_3CH_2CH_2CH_2OH$
propanal	butanoic acid	butan-1-ol
$CH_3CH_2CH_2CHO$	$CH_3CH_2CH_2CH_2NH_2$	
butanal	butan-1-amine	

Refer to page T20.

N8 Examination questions often ask students to compare the boiling points of compounds of different homologous series.
e.g. $CH_3(CH_2)_3OH$ > $CH_3(CH_2)_3Cl$ > $CH_3(CH_2)_3CH_3$ > $CH_3(CH_2)_2CH_3$

29.19 Common names of carbon compounds

The development of a formal system for naming carbon compounds did not come about until near the end of the 19th century. Prior to that time many carbon compounds had already been discovered. The names given to these compounds sometimes reflected a source of the compound. For example, formic acid (i.e. methanoic acid) got its name from the Latin word for ant, *formica*, as the acid was first obtained from red ants (Fig. 29.16).

The older names for carbon compounds are now called **common names** or **trivial names**. Many of these names are still widely used. Therefore it is still necessary to learn the common names for some of the common compounds.



Fig. 29.16 This red wood ant can protect itself by spraying methanoic acid to its attackers

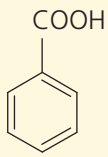
common name 通俗名稱　　trivial name 俗名

Worksheet **Assessment Form**
A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.

Find & Share

Common names of some carbon compounds

Search for the common names of the compounds below and complete the table. Write down the origins of the common names where appropriate. The first one has been done for you.

Structural formula of compound	Common name	Origin of the common name
$HCOOH$	formic acid	from the Latin word for ant, *formica*. (Ants can spray attackers with methanoic acid.)
CH_3COOH	acetic acid	from the Latin word for vinegar, acetum
	benzoic acid	derived from gum benzoin (the resinous product from styrax trees), which was for a long time the only source for benzoic acid.
$HCHO$	formaldehyde	the common name of an aldehyde is often derived from the name of the acid it forms.
CH_3CHO	acetaldehyde	• formaldehyde derived from formic acid; • acetaldehyde derived from acetic acid.
CH_3COCH_3	acetone	common names for ketones can be derived by naming the two alkyl or aryl groups bonded to the carbonyl group as separate words followed by the word ketone. • acetophenone (i.e. phenylethanone) • diethyl ketone (i.e. pentan-3-one)
$CH_3CH(OH)CH_3$	isopropyl alcohol	common names for alcohols usually take the name of the corresponding alkyl group and add the word 'alcohol', e.g. methyl alcohol and ethyl alcohol. Propyl alcohol may be n-propyl alcohol or isopropyl alcohol, depending on whether the hydroxyl group is bonded to the 1st or 2nd carbon of the propane chain.
$CHCl_3$	chloroform	chloroform was named and chemically characterized in 1834 by the French chemist Jean-Baptiste Dumas [Fr. *chloroforme*, from chlor- 'chlorine' + formique 'formic (acid)'].

Have you mastered?

Key terms

functional group 官能基 2 homologous series 同系列 2 prefix 詞首 5 suffix 詞尾 5

haloalkane 鹵烷 8 alcohol 醇 10 hydroxyl group 羥基 10 aldehyde 醛 13

ketone 酮 13 carbonyl group 羰基 13 carboxylic acid 羧酸 15 carboxyl group 羧基 15

ester 酯 17 amide 酰胺 19 amine 胺 20 miscible 互溶 24 dimer 二聚物 27

Checklist

After studying this unit, you should be able to

- ☐ give the IUPAC names, general formulae, condensed formulae and structural formulae for alkanes, alkenes, haloalkanes, alcohols, aldehydes, ketones, carboxylic acids, esters, unsubstituted amides and primary amines;
- ☐ write the structural formulae of the compounds based on their IUPAC names;
- ☐ understand the effects of functional groups and carbon chain length on physical properties of carbon compounds;
- ☐ identify common trivial names of some carbon compounds.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

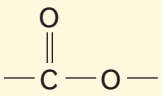
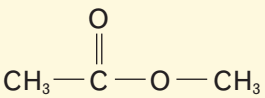
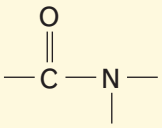
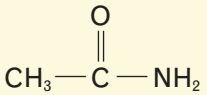
Summary

1 The following table summarizes the naming of compounds in various homologous series.

Homologous series	General formula	Functional group it contains	Naming	Example	
				Structural formula	IUPAC name
Alkanes	C_nH_{2n+2}	—	add appropriate prefix to -ane	$CH_3CH(CH_3)CH_2CH_3$	2-methylbutane
Alkenes	C_nH_{2n}	$>C=C<$	replace 'ane' in the name of the corresponding alkane with -ene	$CH_3CH=CH_2$	propene
Haloalkanes	RX	–X (X = F, Cl, Br or I)	add the name of halogeno functional group as prefix to the name of the corresponding alkane	$CH_3CH(Cl)CH_3$	2-chloropropane
Alcohols	ROH	–OH	replace the last letter 'e' in the name of the corresponding alkane with -ol	CH_3CH_2OH	ethanol
Aldehydes	RCHO	$-C(=O)-H$	replace the last letter 'e' in the name of the corresponding alkane with -al	$CH_3CH_2C(=O)H$	propanal
Ketones	$RCOR_1$	$-C(=O)-$	replace the last letter 'e' in the name of the corresponding alkane with -one	$CH_3C(=O)CH_3$	propanone
Carboxylic acids	RCOOH	$-C(=O)OH$	replace the last letter 'e' in the name of the corresponding alkane with -oic acid	$CH_3CH_2CH_2C(=O)OH$	butanoic acid

Have you mastered?

Continued on next page

Homologous series	General formula	Functional group it contains	Naming	Example	
				Structural formula	IUPAC name
Esters	$RCOOR_1$		the name consists of two separate words, the first word comes from the alcohol, the second word comes from the acid		methyl ethanoate
Amides	$RCONH_2$ (unsubstituted)		replace the 'oic acid' ending in the name of the corresponding acid by -amide		ethanamide
Amines	RNH_2 (primary)		replace the last letter 'e' in the name of the corresponding alkane with -amine	CH_3NH_2	methanamine

2 The physical properties (e.g. the boiling point and water solubility) of a carbon compound are affected by

a) the functional group it contains;

b) the length of the carbon chain in molecules.

Have you mastered?

3 The following table summarizes the physical properties of members of some homologous series.

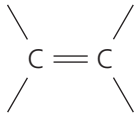
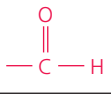
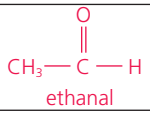
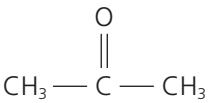
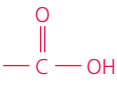
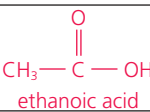
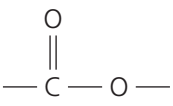
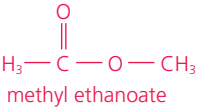
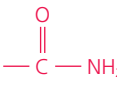
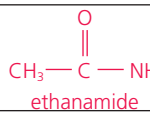
Homologous series	Intermolecular forces	Physical properties
Haloalkanes	• van der Waals' forces between molecules	• boiling points increase with the length of carbon chain
	• polar molecules can interact with water molecules	• slightly soluble in water
Alcohols	• hydrogen bonding between molecules	• boiling points increase with the length of carbon chain • boiling points much higher than those of alkanes of similar relative molecular masses
	• hydrogen bonding between alcohol molecules and water molecules	• alcohols with less carbon atoms are miscible with water in all proportions • alcohols with long carbon chains are much less soluble in water
Aldehydes and ketones	• van der Waals' forces between molecules	• boiling points increase with the length of carbon chain • boiling points lower than those of corresponding alcohols
	• hydrogen bonding between aldehyde / ketone molecules and water molecules	• aldehydes and ketones with small molecules are miscible with water in all proportions
Carboxylic acids	• hydrogen bonding between molecules; more extensive than that in alcohols	• boiling points higher than those of alcohols of similar relative molecular masses
	• hydrogen bonding between acid molecules and water molecules	• the first four acids are miscible with water in all proportions
Esters	• van der Waals' forces between molecules	• boiling points lower than those of alcohols and acids of similar relative molecular masses
	• hydrogen bonding between ester molecules and water molecules	• simple esters are very soluble in water
Amides	• hydrogen bonding between molecules	• boiling points are high
	• hydrogen bonding between amide molecules and water molecules	• simple amides are very soluble in water
Amines	• hydrogen bonding between molecules of primary amines; hydrogen bonding less strong than that in alcohols	• boiling points lower than those of alcohols of similar relative molecular masses
	• hydrogen bonding between primary amine molecules and water molecules	• simple primary amines are very soluble in water

Unit Exercise

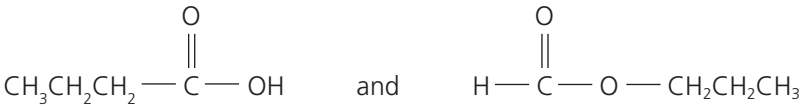
Note: The symbol ! indicates the level of difficulty of a question.

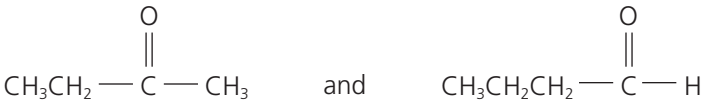
Part I Knowledge and understanding

1 Complete the following table. The first row has been done as an example.

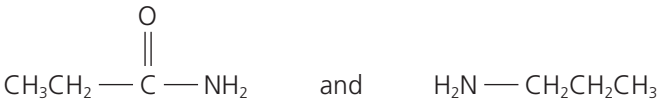
Homologous series		Formula of functional group	Example (structural formula and IUPAC name)
Name	General formula		
Alkenes	C_nH_{2n}		$H_2C=CH_2$ ethene
Haloalkanes	RX	–X	CH_3CH_2Cl chloroethane
Alcohols	ROH	–OH	CH_3CH_2OH ethanol
Aldehydes	RCHO		 ethanal
Ketones	$RCOR_1$		 propanone
Carboxylic acids	(H or R)COOH		 ethanoic acid
Esters	$(H \text{ or } R)COOR_1$		 methyl ethanoate
Amides	$RCONH_2$		 ethanamide
Amines	RNH_2	$-NH_2$	$CH_3CH_2NH_2$ ethanamine

2 Decide whether the compounds in each of the following pairs belong to the same homologous series. Explain your answer in each case.

a)  and

b) 

c) 

d) 

a) No
The first one is a carboxylic acid while the second one is an ester.

b) No
The first one is a ketone while the second one is an aldehyde.

c) Yes
They are both alcohols.

d) No
The first one is an amide while the second one is an amine.

3 Propose the structure of a molecule that meets each of the following descriptions.
Refer to page T21.

a) An alcohol with the molecular formula $C_4H_{10}O$.

b) An alkene with the molecular formula C_6H_{12}.

c) A ketone with the molecular formula $C_5H_{10}O$.

d) An ester with the molecular formula $C_6H_{12}O_2$.

Part II Multiple choice questions

4 Compounds in which of the following pairs belong to the same homologous series?

A $CH_3CH_2OCH_3$ and $(CH_3)_2CHOH$
B $CH_3CH_2CH{=}CH_2$ and $CH_3C{\equiv}CCH_3$
C $CH_3CH_2CH_2COOH$ and $CH_3COOCH_2CH_3$
(D) $CH_3COOC_2H_5$ and $HCOOC_3H_7$

Both $CH_3COOC_2H_5$ and $HCOOC_3H_7$ are esters.

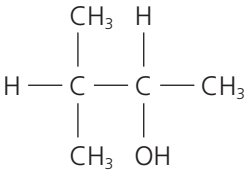
5 Which of the following molecules is a methyl ester?

A $CH_3COOCH_2CH_3$
(B) $HCOOCH_3$
C $CH_3COCH_2CH_3$
D CH_3COCl

$HCOOCH_3$ is methyl methanoate. It is a methyl ester.

(Edexcel Advanced GCE, Unit 4, Jan. 2010, 7)

6 The structure of compound R is shown below:

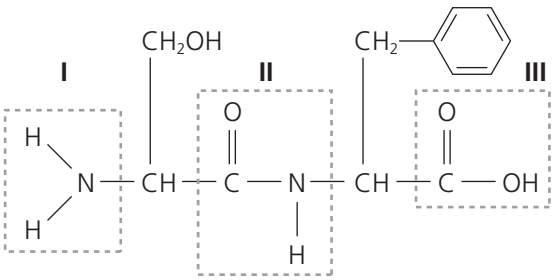


The systematic name of R is

A 2-methylbutan-3-ol.
(B) 3-methylbutan-2-ol.
C 1,1-dimethylpropan-2-ol.
D 3,3-dimethylpropan-2-ol.

(HKCEE, Paper 2, 2010, 12)

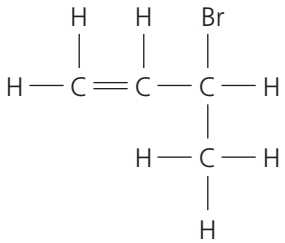
7 The structure of a compound is shown below:



Which of the following combinations about the names of the functional groups labelled I, II and III is correct?

	I	II	III
A	amide	amine	carbonyl
B	amide	ester	carboxyl
C	amine	amide	carboxyl
D	amine	ester	carbonyl

8 What is the systematic name of the following compound?



A 2-bromobut-3-ene
B 3-bromobut-1-ene
C 1-bromo-1-methylpropene
D 3-bromo-3-methylpropene

(HKCEE, Paper 2, 2009, 11)

Directions: Questions 9 and 10 concern the following carbon compounds.

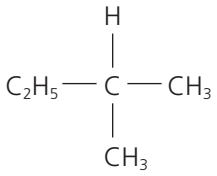
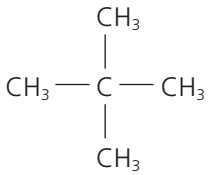
A CH_3CHO
B CH_3COOH
C $CH_3COOCH_2CH_3$
D C_6H_5OH

9 Which compound is most likely to burn with a smoky flame? D C_6H_5OH has a very high carbon to hydrogen ratio. It is most likely to burn with a smoky flame.

10 Which compound is most likely to have a fruity smell? C $CH_3COOCH_2CH_3$ is an ester. It is most likely to have a fruity smell.

(Edexcel Advanced GCE, Unit 5, Jan. 2011, 8(b)–(c))

11 Which of the following alkanes has the highest boiling point?

A $CH_3CH_2CH_2CH_3$
B $CH_3CH_2CH_2CH_2CH_3$
C 
D 

Options B, C and D — Compared with two other compounds ($C_2H_5CH(CH_3)_2$ and $C(CH_3)_4$), the molecular shape of $CH_3CH_2CH_2CH_2CH_3$ allows the greatest surface contact between molecules. The van der Waals' forces in $CH_3CH_2CH_2CH_2CH_3$ is the strongest.

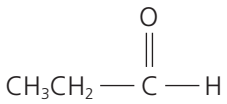
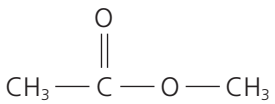
12 Which of the following does NOT have hydrogen bonding in a pure sample, but forms hydrogen bonds with water when it dissolves?

A Propane
B Propanal
C Propanol
D Propanoic acid

Propanal molecules can form hydrogen bonds with water molecules.

(Edexcel Advanced GCE, Unit 4, Jan. 2011, 8)

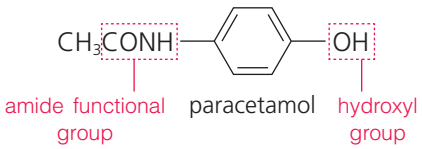
13 Which of the following compounds is likely to be most soluble in water?

A 
B 
C $CH_3CH_2CH_2-O-H$
D $CH_3CH_2CH_2CH_2-O-H$

Options C and D — Molecules of $CH_3CH_2CH_2-O-H$ and $CH_3CH_2CH_2CH_2-O-H$ can form hydrogen bonds with water molecules.

$CH_3CH_2CH_2-O-H$ is more soluble in water than $CH_3CH_2CH_2CH_2-O-H$.

14 The common analgesic drug paracetamol has the structure shown below.

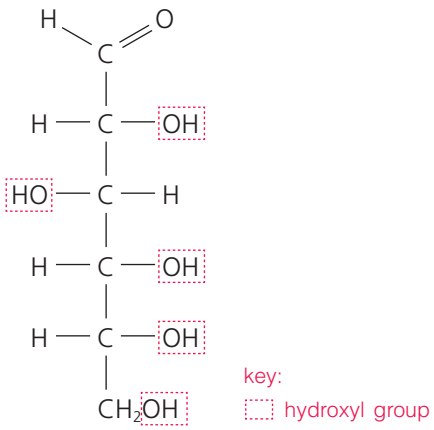


What functional group(s) does the paracetamol molecule contain?

(1) Amide functional group
(2) Amine functional group
(3) Carbonyl group

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

15 The structure of glucose is shown below.



What functional group(s) does the glucose molecule contain?

(1) Carboxyl group
(2) Hydroxyl group
(3) Phenyl group

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

16 Esters can be used to make

(1) perfumes.
(2) food additives.
(3) solvent for paints.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

(HKCEE, Paper 2, 2009, 27)

17 The structure of two carbon compounds are shown below.

$CH_3CH_2CH_2CHO$ (X) $\qquad$ $CH_3COCH_2CH_3$ (Y)

Which of the following statements concerning X and Y is / are correct?

(1) Both contain a carbonyl group.
(2) They belong to the same homologous series.
(3) X is soluble in water while Y is not.

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(2) X is an aldehyde while Y is a ketone. They belong to different homologous series.
(3) Both X and Y are soluble in water.

18 The structures of two carbon compounds, X and Y, are shown below.

$HCOOC_2H_5$ (X) $\qquad$ C_2H_5COOH (Y)

Which of the following statements concerning X and Y is / are correct?

(1) They belong to the same homologous series.
(2) Molecules of both can form hydrogen bonds with water molecules.
(3) The boiling point of X is lower than that of Y.

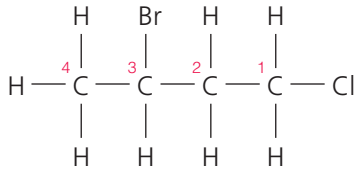
A (1) only
B (2) only
C (1) and (3) only
(D) (2) and (3) only

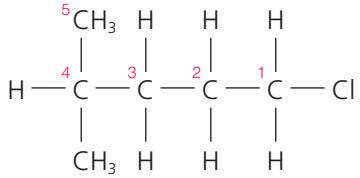
(1) X is an ester while Y is a carboxylic acid. Thus, they belong to different homologous series.
(3) Hydrogen bonds exist in Y but not in X. Thus, the boiling point of X is lower than that of Y.

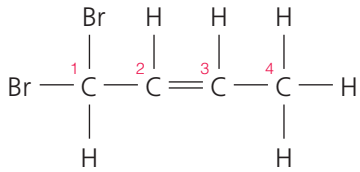
Part III Structured questions

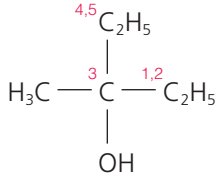
Answers for the questions of the public examinations in Hong Kong are not provided.

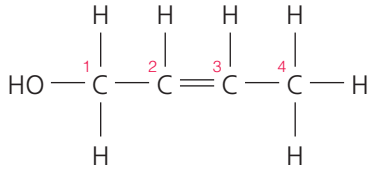
19 Give the IUPAC names of the following compounds.

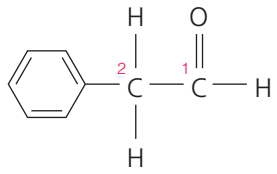
a) 
3-bromo-1-chlorobutane

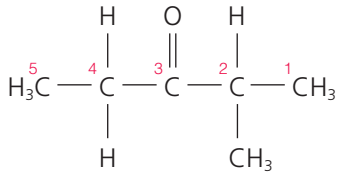
b) 
1-chloro-4-methylpentane

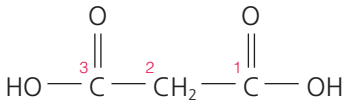
c) 
1,1-dibromobut-2-ene

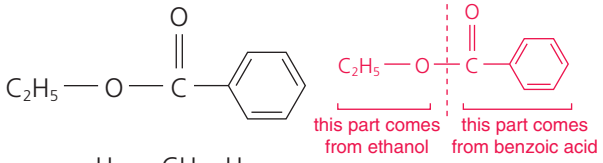
d) 
3-methylpentan-3-ol

e) 
but-2-en-1-ol

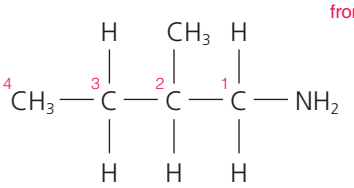
f) 
phenylethanal

g) 
2-methylpentan-3-one

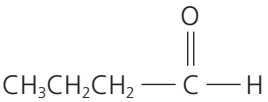
h) 
propanedioic acid

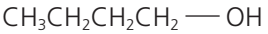
i) 

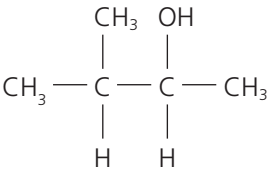
ethyl benzoate

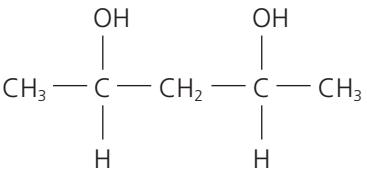
j) 
2-methylbutan-1-amine

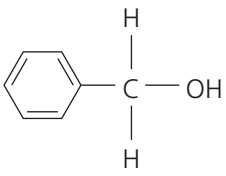
20 Consider the following compounds:

A 

B 

C 

D 

E 

a) Give the IUPAC names of compounds D and E.
D: pentane-2,4-diol E: phenylmethanol

b) Which compound is the most volatile?
A

c) Which compound is the most soluble in water?
D

21 Write the structural formulae of the following compounds.

Refer to page T22.

a) 5-iodopent-2-ene

b) 4-bromobut-3-en-2-ol

c) benzaldehyde

d) pentan-2-one

e) phenylethanoic acid

f) 2-propyl ethanoate

g) ethyl methylpropanoate

h) pentanamide

22 The table below lists the solubility of the first six straight-chain alcohols in water.

Name	Solubility (g per 100 g of water)
Methanol	miscible in all proportions
Ethanol	miscible in all proportions
Propan-1-ol	miscible in all proportions
Butan-1-ol	7.9
Pentan-1-ol	2.3
Hexan-1-ol	0.6

Explain the trend in water solubility of the alcohols as the number of carbon atoms in their molecules increases.

A rule of thumb for predicting solubility is that 'like dissolves like'; polar compounds tend to dissolve in polar solvents.

Methanol, ethanol and propan-1-ol are miscible with water in all proportions. The alkyl groups of these alcohols are relatively small, therefore these alcohols resemble water more than they resemble alkanes.

Alcohols with longer carbon chains are much less soluble in water. This is because these alcohols resemble alkanes more than they resemble water. Van der Waals' forces predominate between these alcohol molecules.

Hence the water solubility of alcohols decreases as the number of carbon atoms in their molecules increases.

23 The boiling points and relative molecular masses of four compounds are listed below.

Refer to page T23.

Compound	Boiling point (°C)	Relative molecular mass
Water H_2O	100	18.0
Ethane CH_3CH_3	–89	30.0
Ethanol CH_3CH_2OH	78	46.0
Propan-1-ol $CH_3CH_2CH_2OH$	97	60.0

Explain why

a) ethanol has a higher boiling point than ethane;

b) water has a higher boiling point than ethanol;

c) propan-1-ol has a higher boiling point than ethanol.

24 Arrange the following compounds in order of increasing boiling point. Explain your answer.

Refer to page T23.

$CH_3(CH_2)_3Br$ $CH_3(CH_2)_3OH$

$CH_3(CH_2)_3Cl$ $CH_3(CH_2)_2COOH$

25 Compound W contains carbon, hydrogen and oxygen only. The relative molecular mass of W is 72.0. Complete combustion of 1.44 g of W gives 3.52 g of carbon dioxide and 1.44 g of water.

Refer to page T24.

a) Deduce the molecular formula of W.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0) C_4H_8O

b) i) Given that W has only one functional group. Draw TWO possible structures of W.

ii) Give the IUPAC name of the compound represented by each structure drawn in (i).